INCH-POUND

MIL-P-44406(GL)
3 May 1990

MILITARY SPECIFICATION

POUCHES, PLASTIC FILM

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers plastic film pouches used for overpacking chemical protective suits.

1.2 Classification. The pouches shall be of one type in the following sizes (see 6.5):

Size 1 - 18-3/4 by 23-1/2 inches
Size 2 - 16-1/4 by 21-1/2 inches

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8105

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SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Untreated
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 882	-	Tensile Properties of Thin Plastic Sheeting
D 1922	-	Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method
D 3951	-	Standard Practice for Commercial Packaging
E 96	-	Water Vapor Transmission of Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Basic material. The basic material for the pouches shall be a colorless, flexible, unsupported film of the following multilayer configuration (see 6.6).

Nylon film (2 mil nominal thickness)
Polyvinylidene dichloride (1/4 mil nominal thickness)
Ethylene vinyl acetate (1-1/2 mil nominal thickness)
Surlyn ionomer (1-1/2 mil nominal thickness)

3.4 Design. The pouches shall be made using the multilayer plastic film (see 3.3.1), heat sealed on three sides as shown in Figures 1 and 2. The pouches shall be made with the Surlyn side to the inside of the pouch. The pouches shall have a vacuum port and tear V notch as specified on Figures 1 and 2.

3.5 Dimensions. The pouches shall conform to the dimensions specified in Figures 1 and 2.

3.6 Physical requirements. The finished pouches shall conform to the physical requirements specified in table I, when tested in accordance with 4.4.3.

TABLE I. Physical requirements

Characteristic	Requirement
Thickness, mils	5 + 1/2, -1/4
Tensile strength, psi, min.:	
Machine direction	3900
Cross machine direction	3500

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TABLE I. Physical requirements (cont'd)

Characteristic	Requirement
Elongation, percent, min.:	
Machine direction	200
Cross machine direction	275
Tear strength, grams:	
Machine direction:	
Minimum	500
Cross machine direction:	
Minimum	200
Maximum	400
Water vapor transmission rate: g/100 sq. in./24 hr, max.	0.5
Heat seal strength, percent of film break strength, min.	45

3.7 Figures. Figures 1 and 2 are furnished solely for guidance and information. Should variation from the written specification appear in the figures, the written specification shall govern.

3.8 Workmanship. The pouches shall be uniformly made, free from holes, tears, cuts, creases, wrinkles or other surface imperfections which might impair their usefulness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies

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submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects listed in 4.4.2 and shall be tested for the characteristics specified in 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

4.4.2 End item visual examination. The pouches shall be examined for the defects listed in table II. The lot size shall be expressed in pouches. The sample unit shall be one pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Material	Any tear, hole or split	101	
	Surlyn side of plastic film not facing to the inside	102	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Material (cont'd)	Not fabricated as specified	103	
	Creases, sharp wrinkles or surface imperfections in body of pouch which impair the use of the pouch	104	
Heat sealing	Not heat sealed as specified	105	
	Not heat sealed on three sides	106	
	Heat seal width not as specified		201
	Heat seal not continuous	107	
V Notch	Notch missing or not as specified		202
	Dimension of tear notch not as specified		203
Vacuum port	Vacuum port missing or not as specified		204
	Dimension of vacuum port not as specified		205
General	Any pouch dimension not as specified	108	

4.4.3 End item testing. The end items shall be tested for the characteristics listed in table III. The lot size shall be expressed in units of pouches. The sample unit shall be two fabricated pouches. Any test failure shall be cause for rejection of the lot. The sample size shall be as follows:

Lot size	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. End item tests 1/

Characteristic	Requirement paragraph	Test Method
Thickness, mils	3.6	ASTM D 882
Tensile strength, psi, min.:		
Machine direction	3.6	ASTM D 882
Cross machine direction	3.6	ASTM D 882

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TABLE III. End item tests 1/ (cont'd)

Characteristic	Requirement paragraph	Test Method
Elongation, percent, min.:		
Machine direction	3.6	ASTM D 882
Cross machine direction	3.6	ASTM D 882
Tear strength, grams:		
Machine direction:		
Minimum	3.6	ASTM D 1922
Cross machine direction:		
Minimum	3.6	ASTM D 1922
Maximum	3.6	ASTM D 1922
Water vapor transmission rate, g/100 sq. in./24 hr, max.	3.6	ASTM E 96 Procedure E
Heat seal strength, percent of film break strength, min.	45	ASTM D 882

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect, illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than specified

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4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2).

5.1.1 Level A packing. Five hundred pouches of one size only, shall be packed flat in a shipping container conforming to style RSC, grade V2s of PPP-B-636. Each fiberboard container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203. In addition, a full size slip sheet of the kraft paper, as specified herein, shall be placed between each one hundred pouches. Each box shall be closed, waterproofed and reinforced in accordance with the appendix of PPP-B-636. Boxes shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.1.2 Level B packing. Five hundred pouches of one size only, shall be packed flat in a shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each fiberboard container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203. In addition, a full size slip sheet of the kraft paper, as specified herein, shall be placed between each one hundred pouches. Each box shall be closed in accordance with Method II, as specified in the appendix of PPP-B-636.

5.1.2.1 Weather-resistant fiberboard shipping containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with the appendix of PPP-B-636.

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5.1.3 Commercial packing. Pouches of one size only shall be packed in accordance with ASTM D 3951.

5.2 Palletization. When specified (see 6.2), pouches of one size only, packed as specified in 5.1.2 and 5.1.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pouches covered by this specification are intended for overpacking chemical protective suits.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3 and 6.3).
e. Levels of packing (see 5.1).
f. Type and class of unit load required (see 5.1.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.1.2.1).
h. When palletization is required (see 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

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6.5 Sizes. The size 1 pouch shall be used to overpack the Suit, Chemical Protective and the size 2 pouch shall be used to overpack the Coat and Trousers, Chemical Protective, Aircrew, Flame Resistant.

6.6 Plastic film. The multilayer plastic film manufactured by Curwood, Inc. (Oshkosh, WI) as Curlon Grade 853 has been tested and found to meet the requirements of 3.3.1.

6.7 Subject term (key word) listing.

AUIB
Chemical protective suit
Overpack

Custodian:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

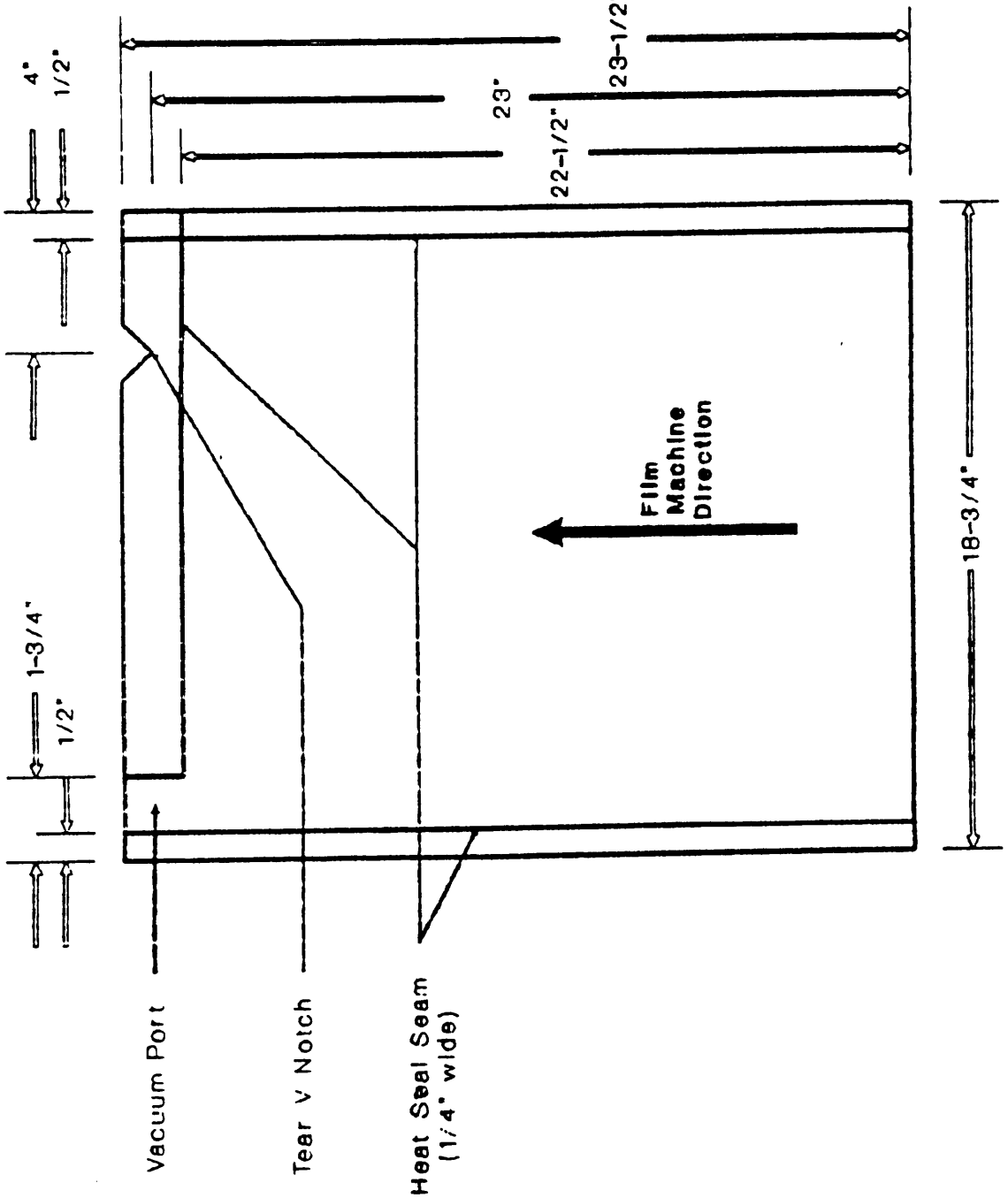
Army - GL

(Project 8105-A331)

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FIGURE 1. POUCH, PLASTIC FILM (SIZE I)

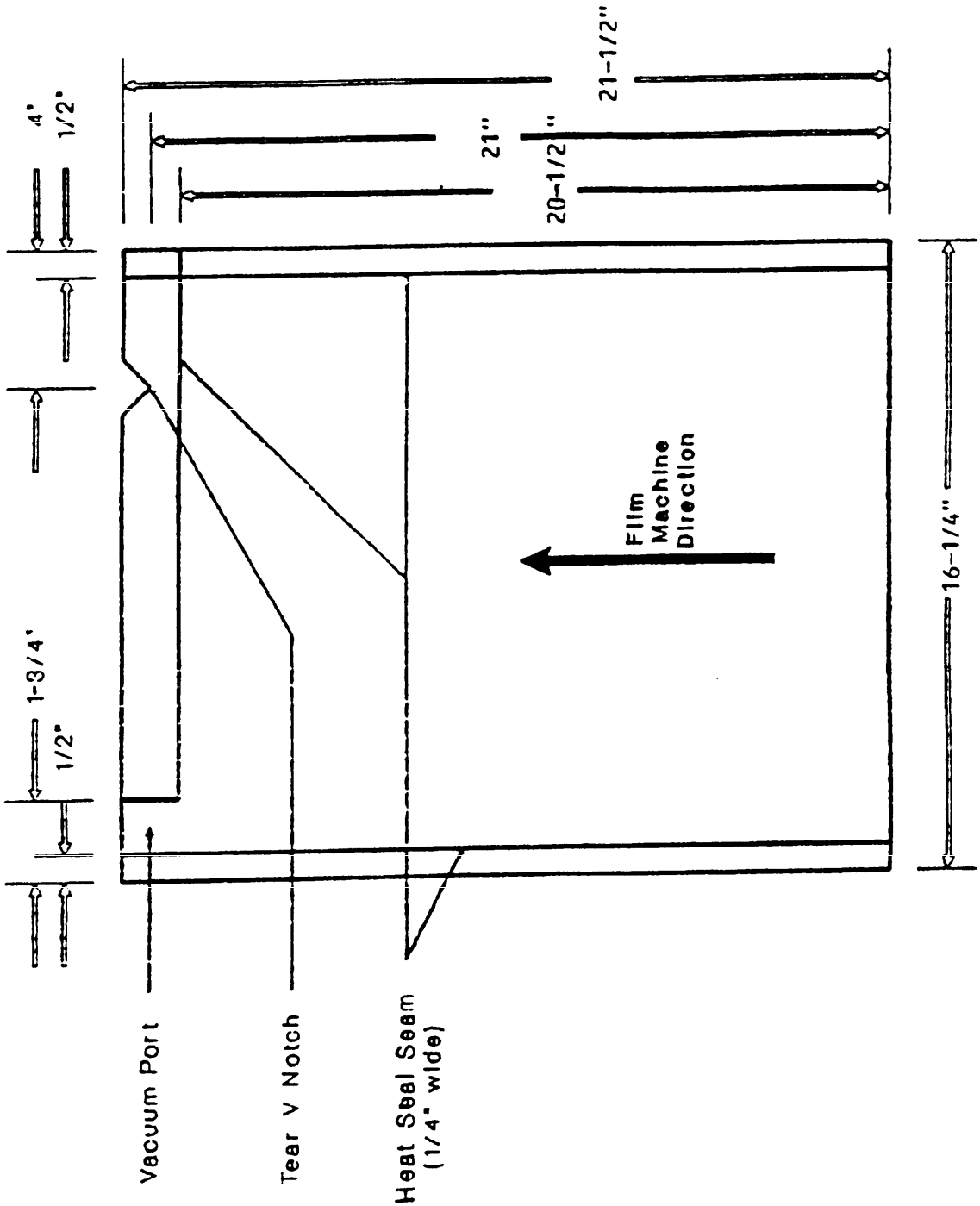
(Nominal Dimensions)



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FIGURE 2. POUCH, PLASTIC FILM (SIZE 2)

(Nominal Dimensions)



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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-44406(GL)	90/05/03

3. DOCUMENT TITLE

POUCHES, PLASTIC FILM

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Natick RD&E Center	508-651-5221	256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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